Expanding a Student's Mind

A Synthetic Observational Study on "Growth Mindsets" within High School Students

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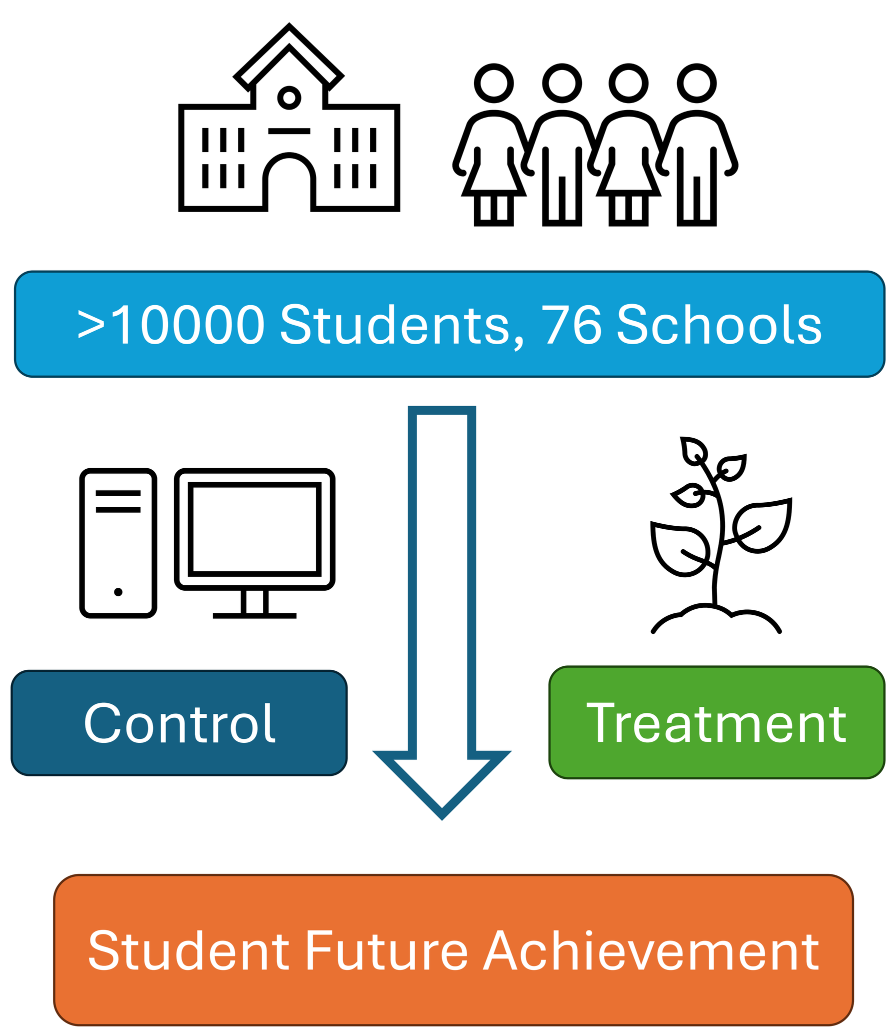
Introduction

Karol Dweck in her book *Mindset: The new Psychology of Success,* outlines two theories on intelligence [1].

Growth Mindset	Fixed Mindset
• Intellectual abilities can be developed	• Intellectual abilities fixed at birth • Immutable

The objective of the National Study of Learning Mindsets (NSLM) experiments was to determine what combination of student and school attributes "benefit most from an online program designed to foster a growth mindset during the transition to high school."[2]



Main Objective: estimate the average treatment effect of growth mindset instruction on student's future achievement using IPW, Hajek, and AIPW estimators

Sub Objective: estimate average treatment effect on high and low poverty concentration schools

Acknowledgements

[1] Stanford Teaching Commons. "Growth Mindset and Enhanced Learning." Standford University. https://teachingcommons.stanford.edu/teaching-guides/foundations-course-design/learning-activities/growth-mindset-and-enhanced-learning (accessed April 21 2025)

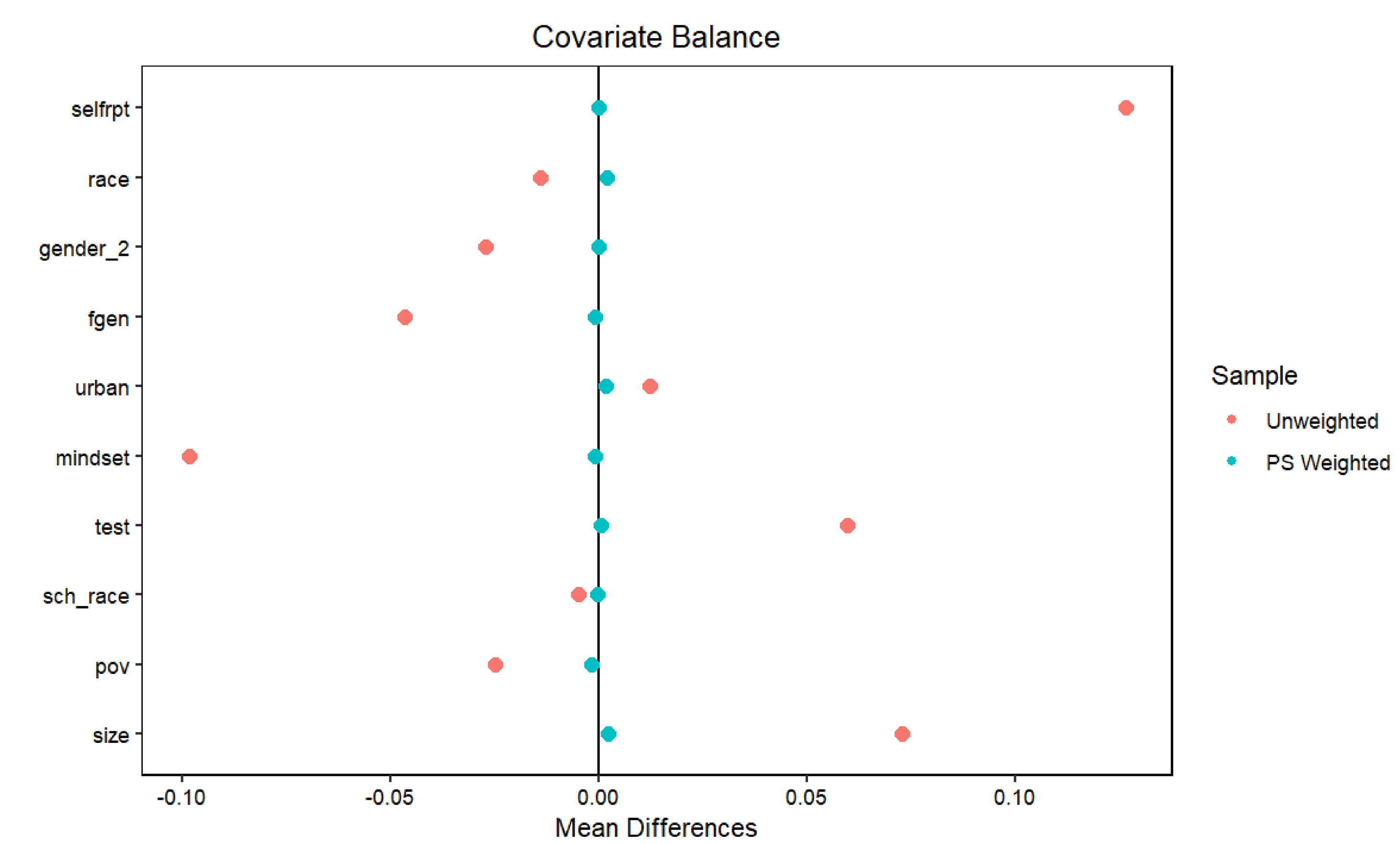
[2] Student Experience Research Network. "National Study of Learning Mindsets." Student Experience Research Network. https://studentexperiencenetwork.org/national-mindset-study/ (accessed April 21 2025)

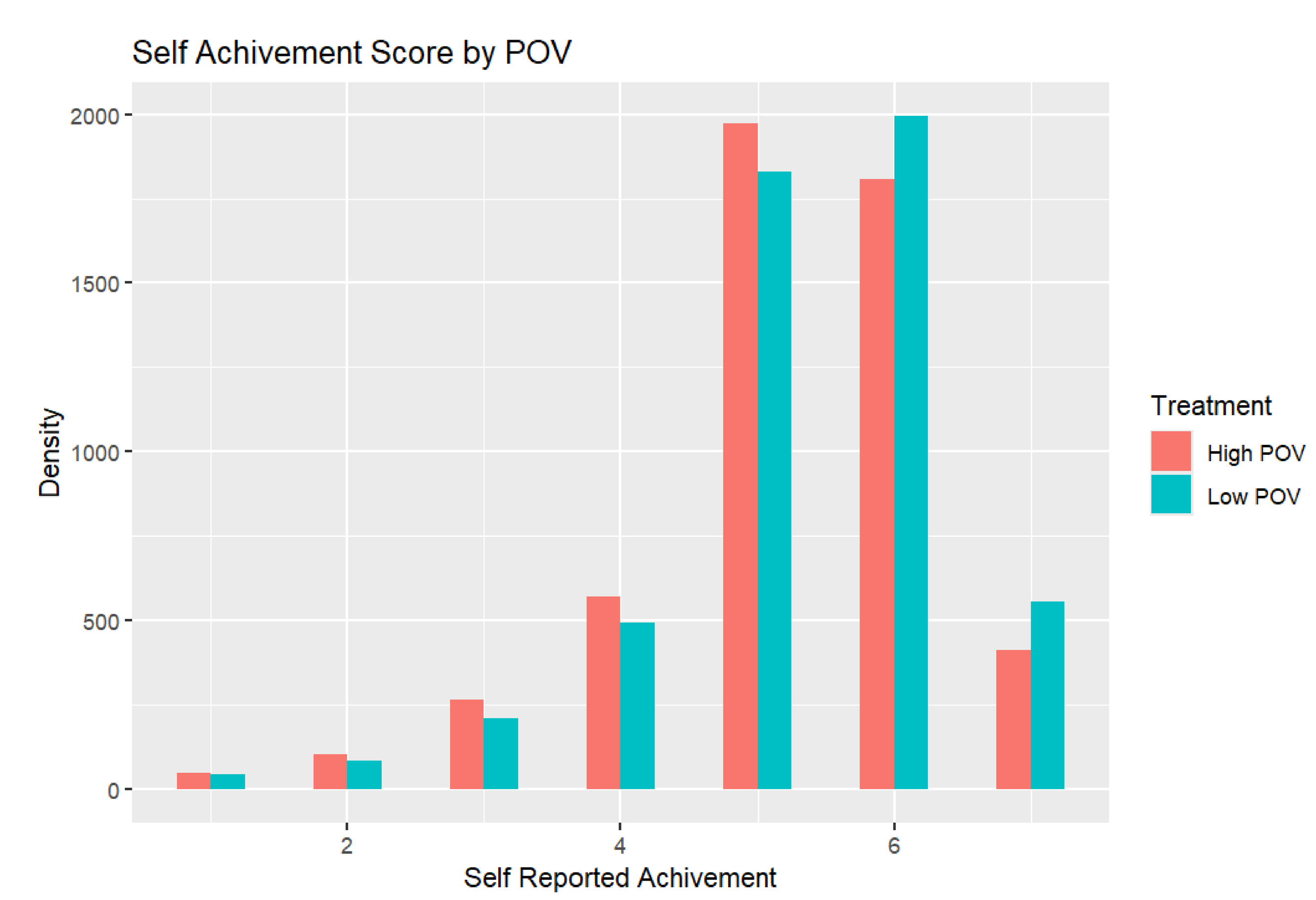
[3] National Center for Education Statistics. "High School Graduation Rates." U.S. Department of Education, Institute of Education Sciences. https://nces.ed.gov/programs/coe/indicator/coi/high-school-graduation-rates (accessed April 21 2025).

Data Exploration

Student Level	School Level
• Self Expectation of Future Success • Race • Gender • First Gen Student	• Urbanicity • Prior student mindsets • Achievement level • Racial Minority Composition • Poverty Concentration • Number of Students

Treatment (z) – growth mindset training
Outcome (y) – achievement in future





Results

Overall Estimated Treatment Effect (95% CI)

$$\boxed{0.386 < \tau < 0.442}$$

High POV Treatment Effect (95% CI) $\boxed{0.499 < \tau < 0.570}$

Low POV Treatment Effect (95% CI) $\boxed{0.259 < \tau < 0.334}$

Data Analysis

Overall

Method	Estimate	Variance	StdError	TestStat	CI_Lower	CI_upper
IPW	0.4141	0.0002	0.0142	29.2377	0.3863	0.4418
Hajek	0.4141	0.0002	0.0142	29.2408	0.3863	0.4419
AIPW	0.4143	0.0002	0.0142	29.2571	0.3865	0.4421

AIPW Hypothesis Test for τ

1. $H_0: \tau = 0, H_A: \tau \neq 0$
2. $\alpha = 0.05$
3. $t = \frac{\hat{\tau}_{AIPW}}{\sqrt{\hat{var}(\hat{\tau}_{AIPW})}} = 29.257$
4. Assuming SUTVA, Strong Ignorability, Overlap, correct Propensity Score and large sample size, t ~ N(0,1)
5. $z_{1-\frac{\alpha}{2}=0.975} = 1.96$
6. Because $|29.257| > 1.96$, we can reject H_0 at 0.05 level, we have evidence that growth mindset instruction leads to greater future achievement

81% of economically disadvantaged students graduate high school in 4 years, compared to the U.S. average at 87%.[3]

High Poverty Concentration

Method	Estimate	Variance	StdError	TestStat	CI_Lower	CI_upper
IPW	0.5342	0.0003	0.0184	29.0038	0.4981	0.5703
Hajek	0.5344	0.0003	0.0184	29.0001	0.4983	0.5705
AIPW	0.5344	0.0003	0.0182	29.2867	0.4986	0.5701

Low Poverty Concentration

Method	Estimate	Variance	StdError	TestStat	CI_Lower	CI_upper
IPW	0.2960	0.0004	0.0189	15.6326	0.2589	0.3331
Hajek	0.2960	0.0004	0.0189	15.6334	0.2589	0.3331
AIPW	0.2969	0.0004	0.0192	15.4911	0.2593	0.3345

Conclusion

- May not have all covariates needed for model
- Synthetic data may not accurately reflect real world

1. There is a positive overall treatment effect of growth mindset instruction on student's future achievement.
2. The growth mindset instruction has a greater positive effect on students in higher poverty schools.